

Milestone 1: Authentication & Authorization

Project: TripNest Travel Platform **Scope:** Backend Security & Frontend Auth Setup

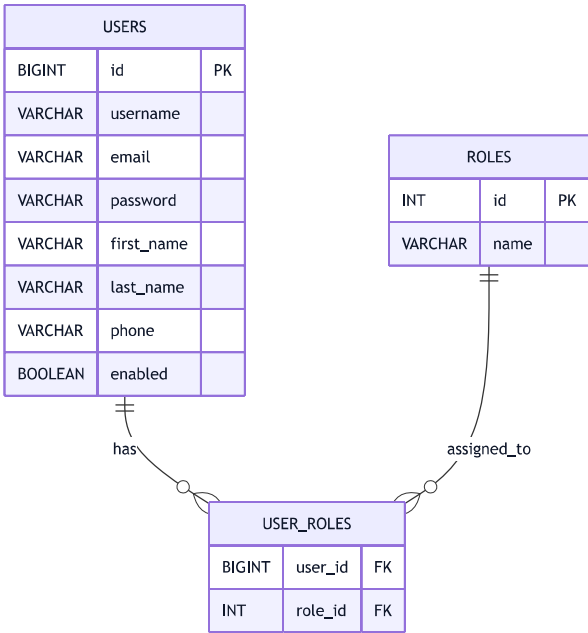
1. System Overview

This report documents the design and implementation details for Milestone 1 of the TripNest application. The core objective of this milestone is to establish a secure, stateless JWT authentication system between the Spring Boot REST backend and the React frontend client.

All endpoints are secured using Spring Security, with role-based routing controls mapped contextually to User roles during runtime.

2. Database Entity Relationship (ER) Diagram

The database schema structure consists of tables managing User credentials, system Roles, and their association mappings to handle RBAC (Role-Based Access Control) requirements.



3. Summary of Completed Operations

Backend Framework (Spring Boot)

- **Database Configuration:** Wired MySQL connectivity properties and configured hibernate properties for automatic updates.
- **Data Seeding:** Set up seed scripts in SQL to initialize required security role privileges (ROLE_TRAVELER , ROLE_AGENT , ROLE_ADMIN).
- **Password Security:** Applied BCrypt encryption hashes to protect stored passwords from cleartext vulnerabilities.
- **JWT Generation & Filters:** Integrated libraries to issue signed cryptographic tokens. Added request filters to parse authorization headers on API calls.
- **Authentication Controller:** Provided signup and signin APIs to process request schemas and generate token payloads.

Frontend Interface (React Client)

- **State Management:** Built an Auth Context provider to maintain user variables and cached security tokens inside local storage.
- **Login Interface:** Created a centered login credential input card.
- **Signup Interface:** Created a centered account creation card that accepts personal details and user role picks.
- **Dashboard Welcome View:** Created a clean screen showing "Welcome, [Name]! Logged in successfully!" and a logout trigger.

4. REST API Endpoint Catalog

Method	Endpoint Path	Access Type	Description
POST	/api/auth/signup	Public	Registers a new User, validates uniqueness, hashes password, and links roles.
POST	/api/auth/signin	Public	Validates user credentials, compiles authorities, and generates JWT.

Evaluation Validation Note

The code builds successfully. The frontend React application and backend Spring Boot server have been launched and tested, confirming successful user registration, login, token distribution, state tracking, and authentication clearing.